Project Report on

AnguliLekha: Real-Time ISL Interpreter with Learning Module

is submitted in partial fulfilment of the requirements of the degree B.E. (Computer Engineering)

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CERTIFICATE

This is to certify that the Project entitled ***"AnguliLekha: Real-Time ISL Interpreter with Learning Module"*** *in the partial fulfilment for* ***BE*** *(Computer Engineering) Degree semester VII during the academic year 2024-25 as prescribed by University of Mumbai.*

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Abstract

The Indian Sign Language (ISL) learning and recognition system presented in this project is designed to support the structured and interactive learning of ISL gestures. Leveraging machine learning and computer vision, the system enables real-time recognition of ISL gestures using standard webcams to facilitate effective learning. The learning process is divided into three progressive levels: alphabets, common words, and everyday phrases or sentences. Users receive immediate feedback, encouraging continuous improvement through an intuitive and user-friendly interface. Gesture recognition is powered by Long Short-Term Memory (LSTM) networks implemented in TensorFlow and Keras, with OpenCV handling video frame processing and feature extraction. This solution promotes accessible ISL education and serves as a valuable tool for learners aiming to acquire proficiency in Indian Sign Language.

***Keywords**: Indian Sign Language (ISL), Gesture Recognition, LSTM, TensorFlow, Keras, OpenCV, Computer Vision, Real-Time Learning, Sign Language Education, Webcam-based Interface*

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List of Abbreviations

Sr. No.	Abbreviation	Full form
1	ISL	Indian Sign Language
2	LSTM	Long Short-Term Memory
3	KNN	K-Nearest Neighbors
4	SVM	Support Vector Machine
5	ANN	Artificial Neural Network
6	HMM	Hidden Markov Models
7	ML	Machine Learning
8	CNN	Convolutional Neural Network
9	RGB	Red Green Blue (Color space)
10	API	Application Programming Interface

1. Introduction

Sign language serves as a vital mode of communication for individuals who are deaf or hard of hearing, enabling the expression of thoughts and emotions through hand gestures, facial expressions, and body movements. While American Sign Language (ASL) has seen significant progress in gesture recognition technologies, Indian Sign Language (ISL) remains comparatively underexplored in both research and application domains [1]. This disparity has limited the development of supportive educational tools for the Indian deaf community, highlighting the urgent need for accessible, ISL-specific learning platforms.

Previous efforts in sign language recognition have utilized diverse approaches, including glove-based systems, vision-based methods, and 3D modeling techniques. While glove-based solutions offer precision, they are often intrusive and uncomfortable for prolonged use. Vision-based systems provide a more natural interface but often struggle with environmental variability and complex gesture dynamics particularly in the case of two-handed ISL signs, where issues like occlusion and inconsistent lighting reduce recognition accuracy [2].

Several machine learning-based ISL recognition systems have been proposed, some achieving classification accuracies of up to 97.5% using SVM and depth-sensing devices [2]. Simpler methods such as Convex Hull and K-Nearest Neighbors (KNN) have also been implemented to reduce cost, though they often fall short when dealing with dynamic gestures and variable lighting conditions, with reported accuracies around 65% [3]. Webcam-based systems using skin filtering and image preprocessing have achieved up to 96.25% accuracy, but their scope is typically limited to static alphabet gestures and heavily reliant on environmental conditions [4]. Multilingual models incorporating neural networks and Hidden Markov Models have also been tested, but they tend to be hardware-intensive and lack adaptability for ISL contexts [5].

2. Literature Survey

2.1. Literature Survey

Sign language recognition systems have evolved significantly, with early research focusing on both glove-based and vision-based techniques. Paper [1] investigates a variety of methods, including traditional image processing, 3D modeling, and artificial intelligence approaches such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Hidden Markov Models (HMM). These methods enable a high degree of customization and reduce dependence on human interpreters, improving accessibility for the deaf and hard-of-hearing community. However, the system struggles with two-handed signs, which are common in Indian Sign Language (ISL), and is highly sensitive to environmental variables such as lighting conditions and complex backgrounds, which can negatively impact accuracy [1].

Paper [2] presents a real-time gesture recognition system utilizing RGB and depth cameras along with SVM classifiers. The inclusion of depth information enhances hand segmentation and improves the system's robustness across varied users. Notably, the system achieves an accuracy rate of up to 97.5% for a predefined set of ISL gestures. Despite its high performance, the system requires specialized hardware such as Kinect, which limits accessibility and portability. Additionally, the gesture vocabulary supported by the system is limited, reducing its applicability for full-scale ISL communication [2].

A more cost-effective approach is detailed in Paper [3], which combines Convex Hull and K-Nearest Neighbors (KNN) algorithms with preprocessing techniques like segmentation and brightness correction. This method is particularly suited for resource-limited settings, such as educational institutions or public outreach programs. While the system is simple and affordable, it suffers from reduced accuracy—around 65%—especially under challenging conditions like poor lighting, hand occlusion, or rapid movement, making it less suitable for robust, real-world use [3].

Paper [4] introduces a webcam-based recognition system using skin color filtering, histogram matching, eigenvalue extraction, and an Euclidean Distance (ED) classifier. This system achieves high recognition accuracy of 96.25% and operates in real time without the need for wearable devices. It offers a more user-friendly experience while keeping costs low. However, its functionality is limited to recognizing static alphabets, and it depends on fixed threshold values for classification, which may require recalibration for different users or environmental conditions [4].

Finally, Paper [5] conducts a comparative study of various recognition techniques, including Neural Networks, HMM, and SVM, applied to both static and dynamic gestures. The system is capable of multilingual sign language recognition, including ISL, and achieves accuracy rates above 90% in several test environments. While the system effectively bridges communication gaps, it is constrained by language-specific design choices, high hardware costs, and performance issues in real-time scenarios, limiting its scalability and widespread adoption in the Indian context [5].

2.2. Summary of Literature Survey

Paper No.	Technologies Used	Advantages	Disadvantages
[1]	Image Processing, AI (ANN, SVM, HMM), Glove & Vision-based methods, 3D models	Improves accessibility, customizable, reduces interpreter need	Complex for two-handed ISL, sensitive to lighting/background
[2]	SVM, Depth & RGB Cameras, Image Processing	High accuracy (97.5%), real-time, versatile	Limited gesture set, needs special hardware
[3]	Convex Hull, KNN, Preprocessing (segmentation, brightness correction)	Cost-effective, flexible, aids communication	Accuracy ~65%, affected by occlusion & lighting
[4]	Skin Filtering, Histogram Matching, Eigenvalues, ED Classifier	High success rate (96.25%), real-time, no gloves	Limited to alphabets, threshold-dependent
[5]	Camera/Sensor Systems, NN, HMM, SVM,	High accuracy (>90%), multilingual,	Language-specific focus, hardware dependency,

	Static/Dynamic Gestures	bridges communication gap	real-time challenges

Table 2.2.1. Summary of Literature Survey

3. Existing System

The machine learning-based Sign Language Recognition (SLR) landscape has seen considerable development, with a variety of approaches being applied to automate gesture interpretation. However, most existing systems prioritize recognition alone and lack structured learning modules. While they often incorporate classifiers such as SVM, CNN, or KNN for gesture identification, many face significant limitations in usability, particularly for Indian Sign Language (ISL).

Current systems struggle with key challenges such as inconsistent gesture recognition in varying lighting conditions, complex backgrounds, and limitations in recognizing both static and dynamic gestures accurately. Moreover, many focus on isolated gestures rather than continuous or multi-level learning experiences. These shortcomings restrict their use in educational contexts, where structured learning progression—from alphabets to words and phrases is crucial.

Real-time responsiveness is another area where existing systems fall short. High latency and limited feedback make them unsuitable for interactive learning or live communication. Additionally, the lack of comprehensive and diverse ISL datasets reduces model accuracy and generalizability. Many existing solutions are tailored to American Sign Language (ASL), leaving ISL underrepresented in global technological development.

Furthermore, very few systems integrate gesture recognition with an intuitive, level-based learning platform. This limits their potential for long-term user engagement and educational growth. To address these gaps, the proposed system focuses on combining real-time recognition with structured ISL learning, offering users a scalable, responsive, and user-friendly environment tailored to the nuances of Indian Sign Language.

4. Limitation of Existing System

Despite significant advancements in Sign Language Recognition (SLR) technologies, existing systems still present several limitations, particularly when applied to Indian Sign Language (ISL). These limitations impact their effectiveness in real-world applications and their utility as educational tools.

Key limitations include:

- **Lack of Structured Learning Pathways**: Most existing ISLR systems focus solely on gesture recognition without offering a guided learning process. They do not provide level-based progression (alphabets - words - sentences), making them unsuitable for educational use.
- **Insufficient Real-Time Feedback**: Many systems either lack real-time recognition capabilities or exhibit latency, making them impractical for live interactions or immediate feedback, which is essential for effective learning.
- **Limited Dataset Diversity for ISL**: There is a significant scarcity of well-annotated, diverse ISL datasets. This affects model training and generalization, especially when recognizing gestures performed at different angles, speeds, or by different users.
- **Complex Gesture Recognition Challenges**: ISL involves intricate, often dual-hand gestures and facial cues. Many current models, especially those trained on simplified or static datasets, fail to capture this complexity accurately.
- **Unfriendly or Rigid Interfaces**: Current solutions rarely offer user-friendly, interactive interfaces suitable for both deaf learners and hearing users. The lack of integration with video communication tools also restricts usability in real-time conversations.

5. Problem Statement and Objectives

5.1. Problem Statement

The lack of structured, interactive, and accessible Indian Sign Language (ISL) learning platforms poses a significant barrier for individuals seeking to understand and communicate using ISL. Existing systems primarily focus on recognition without offering a guided learning experience. Furthermore, they often fail to provide real-time feedback or a user-friendly environment for self-paced education. The absence of scalable, level-based ISL learning tools restricts both deaf and hearing users from effectively acquiring signing skills, thereby widening the communication gap. This lack of intuitive, educational solutions limits the broader adoption of ISL and hinders inclusive communication in society.

5.2. Objectives

- **Create a Multi-Level ISL Learning Platform**: Design a structured system with progressive levels—alphabets, common words, and everyday phrases to guide users in learning ISL step-by-step.
- **Enable Real-Time Gesture Recognition**: Utilize LSTM-based machine learning models with OpenCV and TensorFlow to provide immediate feedback on gesture accuracy.

  Support Interactive Learning: Incorporate real-time recognition into an interactive interface where users can practice, get evaluated, and track their progress.
- **Improve Accessibility and Inclusivity**: Develop a platform that empowers both hearing and deaf users to learn ISL, promoting inclusive communication in daily life.
- **Ensure a Simple and Engaging UI**: Design an intuitive, webcam-based interface that enhances user engagement and ease of use across all age groups.

6. AnguliLekha

6.1. Framework

The proposed Indian Sign Language (ISL) learning and recognition system is developed using a modern web-based technology stack to deliver an interactive and accessible user experience. The system emphasizes real-time gesture recognition, progressive learning modules, and visual feedback to support structured gesture learning.

Core Components of the Framework:

- **Frontend:** The user interface is built using React and styled with CSS, providing a smooth, responsive, and intuitive environment. It allows learners to choose learning levels, view gesture recognition results, and receive visual guidance and feedback.
- **Model Deployment:** The trained ISL recognition model initially created using Google's Teachable Machine is exported and integrated into the frontend using TensorFlow.js. This enables gesture recognition directly in the browser without requiring server-side computation, ensuring privacy and low latency.
- **Learning Modules:** The system is structured into three levels: alphabets, words, and sentences. Each module contains gesture demonstrations, practice interfaces, and progression tracking.
- **Camera Input & Gesture Processing:** The webcam stream is accessed through JavaScript, and each frame is processed in real time using TensorFlow.js to classify the performed gesture based on the trained model.
- **Feedback Mechanism:** Instant feedback is displayed on-screen, guiding learners to correct gestures when necessary. Positive feedback is shown upon correct classification, enhancing engagement and motivation.

6.2. Design Details

The design of the proposed ISL recognition system follows a structured pipeline starting from real-time video capture to gesture classification and feedback. It utilizes OpenCV for frame extraction and preprocessing, followed by an LSTM-based model for recognizing gesture sequences. The system then displays recognized text and guides users through progressive learning modules via an intuitive Python-based interface.

6.2.1. Module For Indian Sign Language Learning Learning Software

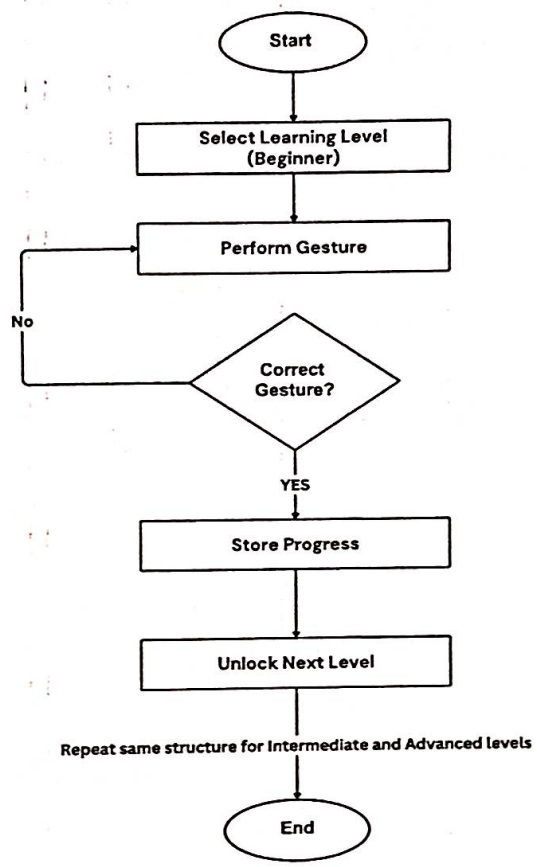


Figure 6.2.1. Module for Indian Sign Language Learning Software.

The learning module guides users through three structured levels:

- **Beginner:** Alphabets
- **Intermediate:** Common Words
- **Advanced:** Sentences and Everyday Phrases

Users begin at the Beginner level and must correctly perform each gesture to unlock subsequent stages. Every attempt is evaluated by the recognition model; correct gestures trigger progression, while incorrect ones result in visual feedback and allow for retries. The system tracks user performance to ensure a personalized and mastery based learning experience.

6.2.2. Module for Indian Sign Language Recognition (ISLR)

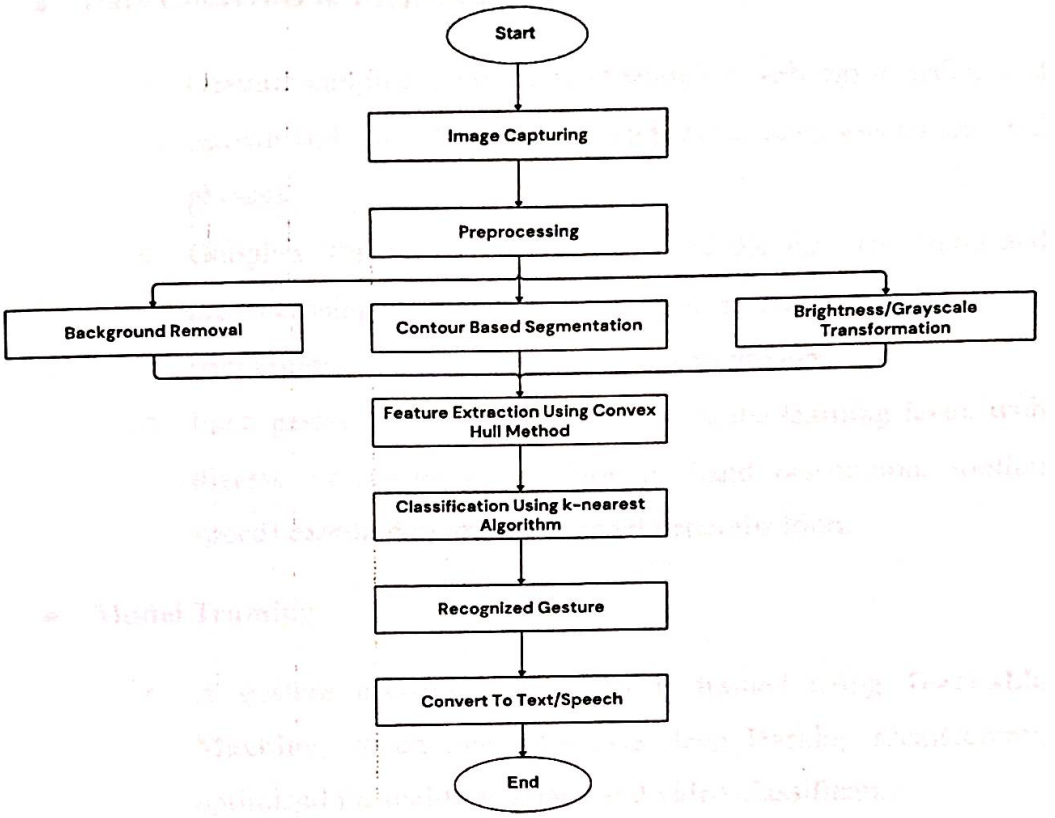


Figure 6.2.2. Module for Indian Sign Language Recognition (ISLR).

The recognition module captures video through the user's webcam and processes individual frames in real time using TensorFlow.js. A pre-trained model, exported from Teachable Machine, is used to classify each gesture based on hand positions and movements. The system displays the recognized gesture along with a confidence score on the interface. If the performed gesture does not match the expected one in the learning module, helpful tips and visual hints are shown to guide correction allow for retries. The system tracks user performance to ensure a personalized and mastery based learning experience.

6.3. Methodology

The methodology for the proposed ISL gesture learning platform involves four core components: data acquisition, model training, real-time recognition, and structured learning progression. The platform leverages browser-based technologies and pre-trained machine learning models to ensure an accessible, interactive user experience.

- **Data Collection & Preprocessing**
    - Gesture samples are collected through a webcam interface and categorized into three levels: alphabets, common words, and phrases.
    - Google's **Teachable Machine** is used for data recording and preprocessing. It handles tasks such as frame normalization, background filtering, and gesture segmentation.
    - Each gesture class is labeled based on the learning level, with diverse variations (e.g., lighting, hand orientation, motion speed) captured to improve model generalization.
- **Model Training**
    - A gesture classification model is trained using **Teachable Machine**, which internally uses deep learning architectures optimized for real-time image and video classification.
    - The trained model is exported in a **TensorFlow.js** format, allowing seamless integration into the frontend.
    - The model learns to recognize hand gestures based on spatial features from the webcam input.

- **Real-Time Gesture Recognition**
    - The live webcam feed is accessed using JavaScript APIs.
    - Video frames are continuously fed into the **TensorFlow.js** model within the browser.
    - The predicted gesture is displayed as on-screen feedback in real time, along with accuracy/confidence indicators.
    - No server-side processing is required, ensuring low latency and increased accessibility.
- **Learning Interface & Level System**
    - Users interact with a React-based interface to select from three levels:
        - Level 1-Alphabets
        - Level 2-Common Words
        - Level 3-Phrases & Sentences
    - The system tracks progress, providing encouraging feedback upon correct gestures and guidance when corrections are needed.
    - Each level is locked until the learner successfully completes the prior one, promoting structured, mastery-based learning.

7. Experimental Setup

7.1. Details of Database

The proposed system uses a structured, multi-level Indian Sign Language (ISL) gesture dataset, organized to support a progressive learning experience:

- **Level 1 – Alphabets:**
  Includes static image samples of ISL alphabet gestures performed by multiple individuals. Variations in hand size, lighting, and orientation ensure robustness in recognition.
- **Level 2 – Common Words:**
  Comprises frequently used ISL signs for everyday vocabulary such as greetings, objects, and actions.
- **Level 3 – Phrases and Sentences:**
  Contains static or short-sequence gesture representations of commonly used phrases to support advanced learners.

Each gesture sample is collected via webcam and labeled using Google's Teachable Machine interface. The dataset emphasizes diversity in signers and conditions to improve generalization. Since the system uses image-based classification (not sequence learning), no LSTM preprocessing is required.

7.2. Performance Evaluation Parameters

The system's performance is evaluated across the following parameters:

- **Recognition Accuracy:**
  Measures the percentage of correctly identified ISL gestures at each learning level. Accuracy is validated through both test samples and live webcam interaction.
- **Real-Time Responsiveness:**
  Evaluated based on how quickly the browser-based system predicts gestures from webcam input. Leveraging **TensorFlow.js**, predictions are generated client-side with minimal delay.
- **Learning Progression & Feedback:**
  Tracks user progress through the structured levels. Real-time feedback helps learners improve gesture accuracy and unlock subsequent levels.
- **User Experience Metrics:**
  Assessed via user testing and surveys focused on usability, visual clarity, and overall learning engagement. Metrics include ease of use, perceived learning value, and satisfaction.

7.3. Software and Hardware Setup:

Hardware:

- **Processor (CPU):** Intel Core i5 (8th Gen) or AMD Ryzen 5 equivalent or higher (for smooth real-time video processing and model inference in browser)
- **RAM:** 8 GB minimum (16 GB recommended for better performance)
- **Webcam:** Standard built-in or external webcam (720p or higher resolution recommended)
- **Storage:** Minimum 10 GB free disk space

Software Stack:

- **Frontend:** Built using React.js and CSS for an interactive, browser-based interface.
- **Model Training:** Gesture models are trained using Google Teachable Machine, which exports directly to TensorFlow.js format.
- **Model Inference:** Implemented using TensorFlow.js in the browser for live gesture recognition.
- **Interaction Logic:** Developed using JavaScript, handling webcam integration and model prediction feedback.

8. Results Analysis

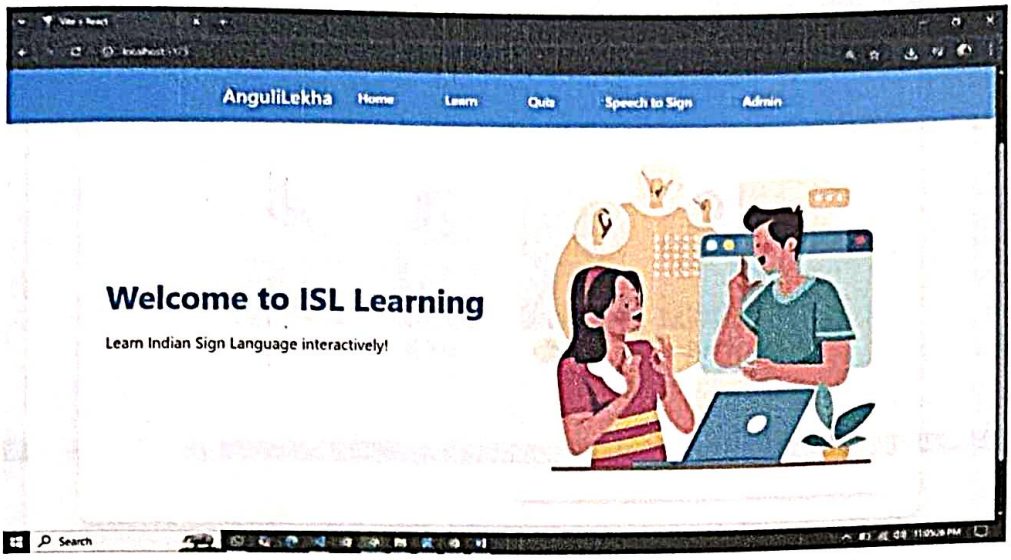


Figure 8.1. Home Screen of AnguliLekha Platform.

The home screen of the AnguliLekha platform offers a welcoming interface that introduces users to various features including Learn, Quiz, Speech to Sign, and Admin sections. The layout is designed to guide users toward exploring ISL in a progressive manner. This screen sets the tone for the platform's educational and accessibility goals, helping learners navigate the system with ease while promoting engagement through its modern, interactive design.

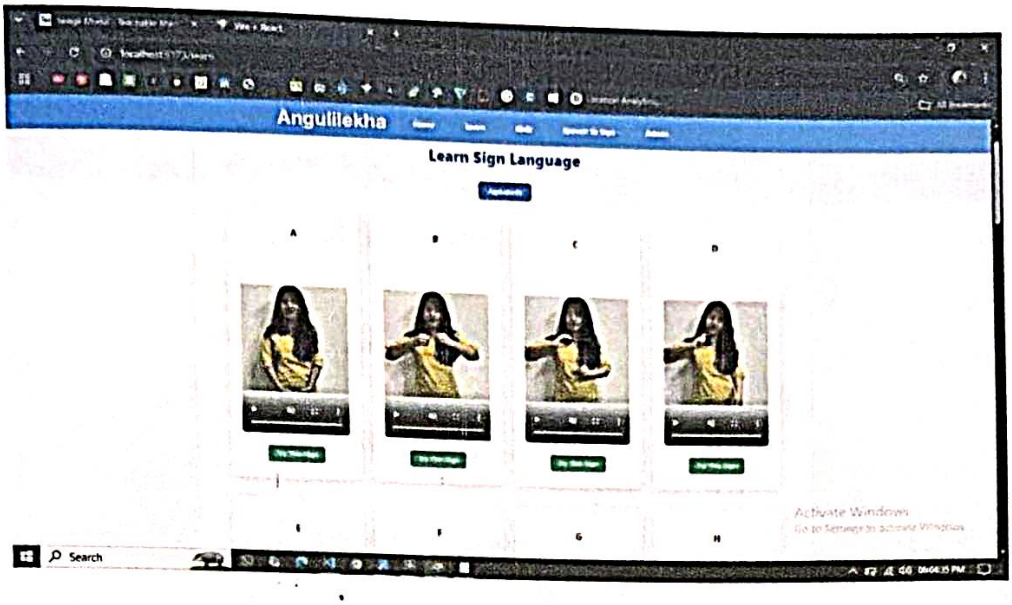


Figure 8.2. Isl-alphabet-learning-module.

ISL learning module of the AnguliLekha platform presents users with a structured layout for exploring the Indian Sign Language alphabet. Each letter is represented with an instructional video and a "Try This Sign" button, encouraging interactive learning. This module enables learners to visually compare their own gestures with standard ISL signs, reinforcing recognition and muscle memory. Such a user-friendly interface bridges the gap between passive content consumption and active practice, making it easier for users to internalize the sign language alphabet.

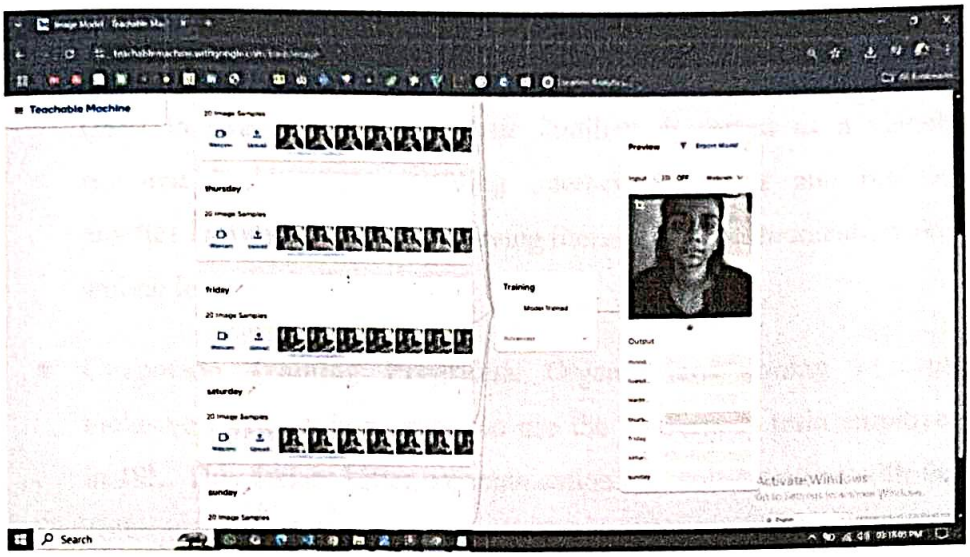


Figure 8.3. Training Phase using Teachable Machine.

The ISL learning module of the AnguliLekha platform presents users with a structured layout for exploring the Indian Sign Language alphabet. Each letter is represented with an instructional video and a "Try This Sign" button, encouraging interactive learning. This module enables learners to visually compare their own gestures with standard ISL signs, reinforcing recognition and muscle memory. Such a user-friendly interface bridges the gap between passive content consumption and active practice, making it easier for users to internalize the sign language alphabet.

9. Applications

- **Educational Platforms for the Deaf Community**: The system can be integrated into educational institutions to teach Indian Sign Language (ISL) to deaf students and their families. It serves as a valuable resource for teachers, enabling interactive lessons and real-time practice sessions, thereby improving literacy and communication skills among learners.
- **Corporate Training Programs**: Organizations aiming to create inclusive work environments can use the platform to train employees in ISL. This fosters better communication and collaboration with deaf colleagues, promoting inclusivity and diversity in the workplace.
- **Healthcare Services**: The platform can be utilized by healthcare professionals to communicate effectively with deaf patients. For example, doctors and nurses can use the system to translate important medical instructions or inquiries into ISL, ensuring clear and accurate communication.
- **Tourism and Hospitality**: Hotels, travel agencies, and tourist centers can deploy the system to assist deaf travelers. By providing a tool for basic ISL communication, the platform ensures a welcoming and accessible experience for tourists with hearing impairments.
- **Public Service Accessibility**: Government agencies and public service providers can use the platform to ensure deaf citizens have equal access to information and services. For instance, it can be used in customer service centers to facilitate effective communication.
- **Personal Learning and Skill Development**: Individuals interested in learning ISL for personal or professional reasons can use the platform to acquire and practice skills at their own pace. Gamified elements and interactive modules make the learning experience engaging and effective.

- **Video Chat Applications**: The system can be integrated into video conferencing platforms to enable real-time communication between deaf and hearing users. The gesture-to-text and text-to-gesture functionality ensures seamless and natural interaction during virtual meetings or social interactions.
- **Research and Development in Gesture Recognition**: The platform's gesture recognition technology can be adapted for other sign languages, advancing computer vision, machine learning, and accessibility solutions.

10. Conclusion and Future Work

10.1. Conclusion

The proposed Indian Sign Language (ISL) learning and recognition system offers a structured, real-time platform for users to effectively learn and practice ISL. By dividing the learning process into three progressive levels—alphabets, common words, and phrases—this system ensures a gradual and comprehensive approach to sign language education. Leveraging LSTM-based models and real-time video processing with OpenCV and TensorFlow, the system delivers instant feedback to enhance the learning experience. With a user-friendly and interactive interface, the platform bridges the communication gap between the hearing and deaf communities, promoting inclusivity, awareness, and accessibility.

10.2. Future Work

- **Expand Dataset Coverage**: Augment the current gesture dataset to include regional sign variations from across India, ensuring better generalization and cultural representation.
- **Enhance Gesture Complexity Handling**: Improve the system's capability to recognize more complex, dynamic, and dual-hand gestures commonly used in conversational ISL.
- **Gamify Learning Modules**: Introduce quizzes, achievements, and progress tracking features to gamify the learning experience and increase user engagement.
- **Mobile and Cross-Platform Support**: Develop lightweight mobile and cross-platform applications to make the system more accessible on various devices.
- **Collaborate with ISL Experts**: Work with certified ISL interpreters and organizations to validate gestures and improve recognition accuracy.

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